Isolating the Safety Function on a Single-Chip Wi-Fi SoC: Formal Verification, Response-Time Analysis, and a Reproducible Measurement Protocol

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Abstract

Consumer intrusion alarms are built almost entirely on single-chip Wi-Fi systems-on-chip, where the radio driver, the network stack and the safety application share one die, one RTOS scheduler and one interrupt controller. The standard engineering mitigation is to pin the safety task to its own core. Behnke et al. have shown that this is not sufficient on an ESP32: under IP-layer load, with the network server already on the other core, a periodic critical task lost 2.15–3.8% of its cycles and lateness grew with packet rate. What remains unknown is whether pinning combined with queue discipline is an adequate *mitigation*, which that work does not evaluate; whether 802.11 management-frame load, handled below the IP stack, behaves differently from IP load; and whether the isolation argument can be established formally rather than empirically. We address all three. We present an ESP32 intrusion alarm whose safety loop is pinned to the application core and separated from the network task by bounded, non-blocking queues and a zero-timeout status snapshot, hardened with HMAC-SHA256 command authentication over a non-volatile monotonic counter. We specify the isolation argument in TLA+ and verify it exhaustively with TLC under a network task given no fairness assumption, and we mutation-test both specifications to show the verified properties are not vacuous. We evaluate the design with 7,200 seeded adversarial trials and 3,000,000 sanitizer-backed fuzzing iterations. An ablation over four command-path designs shows that availability and integrity are separable properties: a rate limiter cannot distinguish a legitimate command from a flood and so sacrifices both, while a static shared token restores availability yet is defeated by capture-and-replay. A parameter sweep of the deauthentication detector revealed that our own initial threshold fired spuriously in 79.9% of benign ten-minute runs, and we revised the firmware accordingly. We are explicit that these results establish *logical* isolation only. Physical timing invariance is a property of the silicon and the closed-source radio driver; we specify the measurement protocol and instrumentation in full, and report no latency figure we have not measured. In its place we contribute two things that reduce the gap: a response-time decomposition proving that the blocking term is identically zero by construction and that the loop's own work is constant-time, leaving a single composite hardware quantity to be measured; and a complete, public validation framework—instrument firmware, wiring, conditions, scripts, pre-registered statistics and falsification criteria—with which any laboratory holding two inexpensive boards can perform and report the measurement. The decomposition also exposed a latent defect in the safety path that no other method we applied could detect.

Keywords: embedded systems security; real-time task isolation; formal verification; TLA+; IoT; ESP32

1 Introduction

A domestic intrusion alarm is a safety-critical system that is bought cheaply and then forgotten. Its single obligation is to produce a loud noise within a bounded time of an intrusion. If an attacker standing outside the building can delay or suppress that noise using inexpensive equipment, the device has failed at the only task it has.

Almost all such devices are now built on single-chip Wi-Fi systems-on-chip. On the ESP32, which dominates this market segment, the Wi-Fi driver, the TCP/IP stack, the MQTT client and the alarm application all execute on one die, under one FreeRTOS scheduler, sharing one interrupt controller. This is a materially different situation from the one studied in most of the real-time literature, where the attacked network interface is a separate device and the protected computation can be placed beyond its reach.

The mitigation practitioners reach for is core pinning: place the safety task on one core and everything network-facing on the other. The advice is reasonable and inexpensive, and it is what the platform makes available. Whether it is sufficient is a separate question, and a partially answered one. Behnke et al. [1] measured a periodic critical task's lateness on an ESP32 under IP packet load with the network server already pinned to the other core, and found measurable degradation: the task lost 2.15–3.8% of its cycles for packets addressed to a listened port and 3.17–6.67% for an unused port, with lateness growing roughly linearly once the microcontroller saturated. Their conclusion was that command-and-control over IP is unsuitable for hard real-time work on such devices.

That finding closes one question and opens three others, which motivate this paper.

First, their work measures the *phenomenon* but does not evaluate a *mitigation*. The follow-up work from the same group proposes two: a priority-aware multiqueue network interface with per-flow interrupt moderation [2], and driver-level early demultiplexing [3]. Neither is available to a developer shipping a stock ESP32. The first was evaluated against a simulated network interface and was never built; the second was prototyped on a Xilinx Zynq-7000, a far larger platform, with measurements taken on a single core. Critically, [3] states that the ESP32's second core is used to separate the observed processes from the test harness rather than as a mitigation, so core pinning is nowhere evaluated as a defence.

Second, all load in that line is applied at the IP layer, as UDP or MODBUS/TCP traffic. The attack that actually threatens a wireless alarm is deauthentication: 802.11 management frames that are processed in the MAC layer and the radio driver, beneath the IP stack, and that the device must handle whether or not any socket is listening. Whether this different interference path behaves like IP load is unknown.

Third, none of this work treats the isolation argument formally, and none models an adversary rather than volumetric stress.

1.1 Contributions

We make the following contributions, each stated with the evidence that supports it and the limits of that evidence.

1. **A deployable isolation design, formally verified.** We combine core pinning with bounded non-blocking queues and a zero-timeout status snapshot, and specify the resulting isolation argument in TLA+. TLC verifies exhaustively that the safety step remains enabled, that the network side never writes alarm state, that the queue stays bounded and that no event is lost without being counted, under a network task given *no fairness assumption* and free to stall indefinitely, flood, fail or crash. Both specifications are mutation-tested (Section 6).

2. **An ablation over command-path designs that separates two properties commonly conflated.** Running one adversary against four designs shows that availability and integrity fail independently: a token-bucket rate limiter cannot distinguish a legitimate command from a flood and so drops both while leaving integrity untouched, and a static shared token restores availability but is defeated the moment one legitimate command is captured (Section 9).

3. **A parameter sensitivity analysis that corrected our own design.** Sweeping the deauthentication detector's threshold and window against a model of benign management-frame traffic showed that our initial threshold of eight fired spuriously in 79.9% of benign ten-minute runs at a realistic busy-

network rate. We revised the firmware default to twelve, which reduces this to 0.25% while still detecting every attack rate swept (Section 9).

4. **A complete, reproducible artifact.** Firmware, specifications, harnesses, analysis scripts and figure sources are public. Results regenerate byte-identically from committed seeds, and continuous integration re-runs the formal verification and a reduced adversarial campaign on every commit.

5. **A response-time decomposition that bounds three of four interference terms by construction.** We decompose the safety loop's worst-case response time into its own computation, blocking, interrupt interference and memory contention, and show that the blocking term is identically zero because the safety loop acquires its only shared resource with a zero timeout, that the loop's own work is constant-time, and that the residual is a single composite hardware quantity. This reduces the open question from four unknowns to one, and identifies exactly what a measurement must isolate (Section 5). The analysis also exposed a latent defect invisible to every other method we applied: an unbounded drain loop in the safety path, now fixed.

6. **A complete, independently executable validation framework for the one claim that requires silicon.** Physical sensor-to-siren latency under radio-layer load cannot be established in simulation. We provide the instrument firmware, wiring, conditions, collection and analysis scripts, pre-registered statistics, and falsification criteria, so that any laboratory with two inexpensive boards can perform and report the measurement (Section 10). We report no measured latency, because we have not measured one.

1.2 Scope and honesty of claims

This paper establishes *logical* isolation: no control- or data-flow dependency allows network activity to block, delay or corrupt the safety path, and no forged, stale or malformed command is accepted. It does *not* establish *timing* isolation, which concerns contention for shared silicon and is a property of the implementation rather than the design. Given [1], we regard it as likely that some timing interference exists; the open question is its magnitude relative to the deadline margin, and whether queue discipline bounds it. We state this distinction wherever a result could be read more strongly than the evidence supports.

2 Background and Related Work

2.1 Real-time degradation from on-chip network activity

The closest work to ours is the line from Behnke and colleagues, summarised in Table 1. Their 2020 study [1] is the measurement that motivates this paper, and is discussed in Section 1. Their multiqueue network interface design [2] maps incoming flows to per-process queues with interrupt moderation, dropping packets for unregistered flows before an interrupt fires; at 5,000 packets/s it prevented roughly 93% of interrupts and reduced the critical task's additional runtime by about 80%. The interface was simulated rather than fabricated, and the paper does not establish whether ESP32-class radio hardware could support per-queue steering. Their journal treatment [3] adds a software route, performing early priority-based demultiplexing inside the driver and achieving a 7.5-fold reduction in per-packet processing for low-priority flows on a Zynq-7000.

2.2 Timing isolation on larger platforms

Outside this device class the finding is consistent: spatial separation combined with a resource regulator preserves real-time behaviour, and the platforms are large. Fischer et al. [4] measure control-task jitter on Linux-class industrial devices under wired floods and find core pinning effective, with a separate co-processor best; the attacked interface there is a physically separate NIC. Meszlényi et al. [5] hold

Table 1: Positioning against the closest work. Deadline degradation on this chip family has already been measured; what remains open is whether the only mitigation available on stock hardware is sufficient, under load that reaches the radio rather than the IP stack.

Work	Platform	Deadline measured	Radio load	Pinning as mitigation	Adversarial
Behnke et al. [1]	ESP32	yes	no	no	no
Behnke et al. [2]	Simulated NIC	yes	no	no	no
Behnke et al. [3]	Zynq-7000	yes	no	no	no
Fischer et al. [4]	Linux industrial	yes	no	no	yes
Meszlényi et al. [5]	x86 PREEMPT-RT	yes	no	no	no
Hoornaert et al. [6]	A53 + hypervisor	yes	no	no	no
Carbajal et al. [10]	Wi-Fi testbed	no	yes	no	yes
Kremser et al. [8]	ESP32-S3	no	yes	no	yes
Cayre et al. [9]	ESP32 family	no	yes	no	yes
This work	**ESP32**	**pending**[†]	**yes**	**yes**	**yes**

[†]Logical isolation is established here by model checking and simulation. The physical deadline measurement is specified in Section 10 but has not been performed, and no latency figure is claimed.

DDS latency predictable under heavy interference using priority-based middleware scheduling, AF_XDP kernel bypass and dedicated NIC hardware queues on x86 with a PREEMPT-RT kernel. Hoornaert et al. [6] co-design CPU-time and memory-bandwidth isolation on a Cortex-A53 MPSoC, switching the interconnect from fair to fixed-priority arbitration on overload using a hypervisor and an FPGA-hosted memory scheduler. The mixed-criticality scheduling literature these sit within is surveyed by Burns and Davis [7], which treats criticality under benign timing assumptions rather than adversarial ones.

None of the mechanisms these works depend upon—a memory management unit, a hypervisor, performance-counter regulation, hardware queues—exists on a 240 MHz microcontroller. What remains available is core pinning and queue discipline, which is precisely why their sufficiency matters.

2.3 Radio-layer attacks on this device class

Deauthentication attacks against the ESP32 family are inexpensive and demonstrated. Kremser and Graffi [8] build deauthentication firmware for the ESP32-S3 itself, patching the compiled image because the vendor transmit path rejects such frames, and note reduced effectiveness against networks using protected management frames. Cayre et al. [9] reverse-engineer the wireless ROM components of the ESP32 family and inject, alter and jam link-layer traffic from software. Carbajal and Jodeiri Akbarfam [10] quantify deauthentication resilience across open, WPA2, PMF and WPA3 configurations. In all of this the dependent variable is whether the *link* survives; none measures whether the victim's *application* continues to meet its deadline. Mudraje et al. [11] reverse-engineer the closed-source ESP32-C3 Wi-Fi driver for static worst-case *energy* analysis, confirming that the driver is opaque and analysable only with substantial effort, but do not address timing or task interference.

2.4 Security and verification on the ESP32 application stack

Work on this platform measures the cost of MQTT over TLS on an ESP32-C3 [12] and runs machine-learning intrusion detection on-device [13]. Both optimise transport cost or detection quality rather than treating a safety deadline as the dependent variable, and an on-device detector is itself additional load on the contended chip. On-device deauthentication detection has been demonstrated previously, and we therefore claim no novelty for our detector; our contribution regarding it is the sensitivity analysis in Section 9. Krichen's survey of formal verification for the Internet of Things [14] covers model checking, abstract interpretation and deductive verification with Event-B, Maude and Simulink tooling, and records no use of TLA+ [15] for this device class.

3 System and Threat Model

3.1 System model

The device is a dual-core microcontroller with a single integrated radio. We write C_s for the core hosting the safety function and C_n for the core hosting all network activity. The safety function is a periodic loop that samples sensors, advances a finite state machine and drives outputs. Let t_d be the instant a physical intrusion asserts a sensor line and t_a the instant the siren is driven. The quantity of interest is the sensor-to-siren latency

$$L = t_a - t_d, \tag{1}$$

and the safety requirement is that $L \leq L_{\max}$ for a deadline $L_{\max}$ fixed by the application.

Let Λ denote the offered load on the radio, in frames per second. Write F_Λ for the distribution of L under load Λ. The isolation property we wish to establish is

$$F_\Lambda \overset{d}{=} F_0 \quad \text{for all } \Lambda \text{ in the operating range,} \tag{2}$$

that is, the latency distribution under load is indistinguishable from the unloaded control. We decompose this into two claims that require different evidence:

Logical isolation. No control- or data-flow dependency exists by which network activity can block, delay or corrupt the safety path. This is a property of the program and is decidable by model checking and simulation.

Timing isolation. Contention for shared silicon—bus, cache, interrupt controller—does not shift F_Λ beyond the deadline margin. This is a property of the implementation on particular hardware and is decidable only by measurement.

Logical isolation is necessary but not sufficient for (2). This paper establishes the former and specifies the experiment for the latter.

3.2 Threat model

The adversary is within radio range, does not possess the device's shared key, and cannot physically access the device. The adversary may: transmit arbitrary 802.11 frames, including deauthentication and disassociation management frames; observe all traffic on the wireless medium, and therefore capture any message the device or its operator transmits; publish arbitrary messages to the device's command topic, either by reaching a shared broker or by operating one; and sustain any of these indefinitely.

Out of scope: physical tampering, side-channel key extraction, supply-chain compromise of the firmware image, and wideband jamming of the 2.4 GHz band, which defeats every wireless channel including the out-of-band fallback and against which the local siren is the only remaining defence. We note this limitation rather than claim coverage we do not have.

The assets to be protected, in priority order, are: the integrity of the alarm state, meaning that no unauthorised party can disarm the system; the timeliness of the local alarm response, meaning $L \leq L_{\max}$; and the delivery of notifications, which is best-effort by design.

4 Design

4.1 Domain separation

The design places the safety function on C_s and all network activity on C_n, and permits exactly three channels between them. Figure 1 shows the arrangement.

Three mechanisms enforce the separation, and each addresses a distinct failure mode.

Core pinning prevents the network task from occupying the safety core's execution slot. The network task is created with an explicit core affinity; the safety loop runs as the default application task on the other core.

Bounded queues prevent either side from exhausting the other's memory or blocking it. The event queue (safety to network) holds sixteen entries and the command queue (network to safety) holds four. On overflow the oldest event is discarded *and counted*, so loss is observable rather than silent; the published device state carries the drop count.

A zero-timeout status snapshot prevents priority inversion from reaching the safety side. The asymmetry here is the mechanism and is easy to overlook: the safety loop acquires the status mutex with a timeout of zero and simply skips the update if the network task holds it, while the network task may wait up to 10 ms. Contention can therefore delay the network side and never the safety side.

4.2 Command authentication

Remote commands carry an authentication tag and a counter. With shared key K and message $m = \texttt{VERB} \,\|\, \text{":"} \,\|\, c$ for counter c, the transmitted tag is

$$\tau = \mathrm{trunc}_{16}(\text{HMAC-SHA256}(K, m)) \,, \tag{3}$$

where HMAC follows RFC 2104 and truncation to 16 octets is permitted by its Section 5, giving a forgery probability of 2^{-128} per attempt. A command is accepted iff

$$\mathrm{ct_eq}\big(\tau, \tau'\big) \;\wedge\; c > c_{\text{stored}}, \tag{4}$$

where τ' is the recomputed tag, $\mathrm{ct_eq}$ is a constant-time comparison that accumulates differences rather than returning early, and c_{stored} is the counter held in non-volatile storage. Both conjuncts are necessary and they defend different things: the first defeats forgery, the second defeats replay. Section 9 shows empirically that a design with only the first is broken by capture-and-replay.

Two placement decisions matter more than the construction, which is standard.

The expensive verification runs on C_n, before the command queue. A flood of forged commands is therefore rejected entirely within the network domain and consumes no safety-core time; only a verdict and an enumerated verb cross the boundary. Figure 2 shows the path.

The counter is committed to non-volatile storage *before* the command is acted upon. Were the order reversed, an interruption between action and commit would leave the command replayable. In the chosen order the worst case is that a counter value is consumed without the command executing, which fails safe: a legitimate sender increments and retries, while an adversary gains nothing.

4.3 Supporting mechanisms

A passive detector counts 802.11 deauthentication and disassociation frames in a sliding window and raises an event when the count reaches a threshold, subject to a cooldown that bounds the report rate. Writing $N(t)$ for the number of such frames in $(t - W,\, t]$, the detector reports at t iff

$$N(t) \geq T \;\wedge\; t - t_{\text{last}} > C. \tag{5}$$

The detector never transmits. The cooldown is not cosmetic: without it the detector could itself flood the bounded event queue, converting a radio attack into a denial of service against the notification path. Parameters T and W were chosen by the sweep in Section 9, which overturned our initial choice.

When the broker is unreachable the device can send a signed, counter-protected alert to a second device over ESP-NOW, which operates without an access point. Optional sensor fusion requires agreement between a passive infrared sensor and a millimetre-wave radar within a window W_f, reducing single-sensor false alarms. Armed state and the command counter persist in non-volatile storage, so a power interruption while armed does not silently disarm the system.

5 Response-Time Analysis

Measurement tells us whether interference occurs; it does not tell us which mechanism produced it, nor what a different design would do. A static decomposition answers the complementary question: of the interference a safety task on this architecture can suffer, how much does this design remove by construction, and how much is irreducibly a property of the silicon?

5.1 Decomposition

A confirmed detection requires the sensor line asserted for T_{conf}, after which the next loop pass advances the state machine and drives the output. Writing R for the worst-case response time of one loop pass,

$$L \;\le\; T_{\mathrm{conf}} + 2R, \tag{6}$$

since at most one pass observes the confirmed edge and at most one further pass drives the output. The quantity to bound is therefore R, which decomposes as

$$R \;=\; C_{\mathrm{own}} \;+\; B \;+\; I_{\mathrm{isr}} \;+\; I_{\mathrm{mem}}, \tag{7}$$

where C_{own} is the loop's uncontended computation, B is blocking on resources shared with the network task, I_{isr} is preemption by interrupt handlers executing on the safety core, and I_{mem} is stall cycles from contention for the shared bus, cache and flash.

5.2 Three terms settled by construction

$B = 0$. The only resource the safety loop shares with the network task is the status snapshot. The safety loop acquires it with a timeout of zero, so the call cannot block: it either succeeds immediately or returns, deferring the update to the next pass. The network task, by contrast, waits up to 10 ms. Therefore $B = 0$ unconditionally, independently of what the network task does, including holding the mutex indefinitely. Priority inversion can delay the network side and cannot delay the safety side. This is precisely the property SAFETYCANALWAYSSTEP verified in Section 6. The cost is that a status snapshot may be skipped, which is acceptable because the snapshot feeds notification, which is best-effort, while the siren is driven from state the safety loop owns outright.

C_{own} is constant-time. Every stage of the loop performs $O(1)$ work. Neither queue can block, both being accessed with a zero timeout; the command drain is bounded by the queue capacity; the display stage, which dominates C_{own} because it performs an I^2C transaction, is rate-limited to once per 250 ms. The deauthentication detector could in principle emit one event per observed frame; it does not, because the cooldown in (5) admits at most one report per 10 s. Without that cooldown a radio flood would become an event flood and the bound would fail, so the cooldown is a timing mechanism as much as a usability one.

The residual. I_{isr} and I_{mem} cannot be bounded from source. The Wi-Fi driver is closed-source; Mudraje et al. [11] required substantial reverse engineering to model its resource consumption for worst-case *energy* analysis and did not address timing. Interrupt core affinity is set by the vendor SDK and not guaranteed, and the memory subsystem's contention behaviour is not published. Hence

$$R \;=\; \underbrace{C_{\mathrm{own}}}_{O(1)} \;+\; \underbrace{0}_{B} \;+\; \underbrace{I_{\mathrm{isr}} + I_{\mathrm{mem}}}_{\text{hardware, unmeasured}} \quad . \tag{8}$$

Three of four terms are settled. The residual is a single composite hardware quantity, and the experiment of Section 10 measures exactly it: the difference between the unloaded and loaded latency distributions isolates $I_{\mathrm{isr}} + I_{\mathrm{mem}}$, because every other term is held constant by construction. Without this decomposition a measured shift could be attributed to any of four causes; with it, only one remains. This is what makes the pending measurement interpretable rather than merely suggestive.

5.3 A falsifiable prediction

We state a prediction before measurement so that it cannot be adjusted afterwards. Given $B = 0$, constant-time C_{own}, and the single-digit-percent cycle loss Behnke et al. [1] measured for IP-layer load, we predict a median latency shift under 10 ms against a T_{conf} of 150 ms, that is under 7% of the confirmation window, with no missed detections. The prediction is refuted by a median shift above 10 ms, by a heavy upper tail indicating occasional interrupt storms even where the median holds, or by any missed detection.

5.4 A defect the analysis exposed

Establishing that every stage is constant-time required checking each one. One was not. The command queue was drained with an unbounded loop, so the safety loop had no static bound on its own execution time: the network core can enqueue while the safety core dequeues, and termination rested on the drain rate exceeding the fill rate. That argument is probabilistic, and weak in the default build, where enqueueing costs only a parse with no tag verification to throttle it. The asymmetry was also inverted, since the network task already bounded its own event drain while the safety task did not. We bounded the drain to the queue capacity, so a full queue still clears in one pass and later arrivals wait at most one loop period.

We record this because of what it says about method. The defect was invisible to every other technique applied here: the test suite passed, the fuzzer found nothing, the adversarial campaign reported no violation, and the TLA+ models have no notion of execution time and so could not have caught it. It was found by asking what bounds each stage, and finding that one stage had no answer.

6 Formal Verification

We specify two models in TLA+ [15] and check them exhaustively with TLC.

6.1 State machine and command path

The first specification models the five alarm states, the transitions between them and the command rules. Over all 25 reachable states at the configured counter bound, TLC verifies that no unauthenticated remote command disarms the system, that the replay counter never decreases, that the siren is never active while disarmed, and that the entry delay and the alarm always resolve rather than deadlocking.

6.2 The isolation argument

The second specification is the one that carries the paper's design claim. It models both tasks, with the network task made maximally adversarial: it may become busy, stall for an unbounded time, fail to publish, crash and restart, and it is granted *no fairness*. Weak fairness is asserted for the safety step alone:

$$\mathit{Spec} \triangleq \mathit{Init} \wedge \Box[\mathit{Next}]_{\mathit{vars}} \wedge \mathrm{WF}_{\mathit{vars}}(\mathit{SafetyStep}). \tag{9}$$

Table 2 lists the verified properties. TLC explored 2,686 states, 660 of them distinct, to a graph depth of 26, with no violation.

6.3 Mutation testing

A model checker reports no violation for a vacuous specification as readily as for a sound one. We therefore injected defects corresponding to each property and confirmed that TLC reports the expected violation: permitting an unauthenticated disarm violates NoUnauthRemoteDisarm, and allowing the network task to write alarm state violates NoNetworkWrite. The properties therefore have force.

Table 2: Properties verified by TLC. Both specifications were additionally mutation-tested.

Property	Type	Statement
NoUnauthRemoteDisarm	invariant	No unauthenticated remote command causes a transition to the disarmed state
CounterMonotonic	property	The replay counter never decreases
NoAlarmWhenDisarmed	invariant	The siren is never active in the disarmed state
SafetyCanAlwaysStep	invariant	The safety step is enabled whenever work remains
NoNetworkWrite	invariant	The network task never writes alarm state
QueueBounded	invariant	The event queue never exceeds its capacity
NoSilentLoss	invariant	produced = delivered + queued + dropped
EntryLeadsToResolve	liveness	An entry delay always ends in alarm or disarm
SafetyProgress	liveness	The safety function makes progress

6.4 What the models do not cover

The abstraction boundary deserves explicit statement, because it is exactly where the research question lives. The network task is modelled as a set of discrete actions, not as a source of *timing interference*. TLC establishes that within this abstraction no control- or data-flow dependency can block the safety step. It says nothing about contention for shared silicon, which is a property of the implementation rather than the model. Model checking here bounds the logical design space; it does not substitute for measurement.

7 Implementation

The firmware comprises approximately 1,100 lines of C++ for the main application, together with four self-contained headers. These provide a portable SHA-256 and HMAC-SHA256 implementation, checked against the published RFC 4231 test vectors; the deauthentication detector; the signed out-of-band alert message; and the sensor-fusion logic. The three latter headers contain pure logic with no platform dependency, which allows them to be tested on a host without hardware.

7.1 Verification infrastructure

The unmodified firmware compiles against host models of Arduino, FreeRTOS, Wi-Fi, MQTT, non-volatile storage and the display. The firmware source is not altered for testing, which is what makes the simulation results meaningful. The suite runs nineteen scenarios at three clock start values across two vendor framework versions, plus build-configuration and header-synchrony checks, for 120 checks in total.

The three clock start values include one immediately below the 32-bit wrap of the millisecond counter. All elapsed-time comparisons are written in the wrap-safe form $(\texttt{uint32_t})(now - since) \geq interval$ rather than $now \geq since + interval$, which fails at the boundary. This is not a theoretical concern: a defect of exactly this class, under which the siren could remain active after approximately 25 days of uptime, was found by this method during development.

7.2 Resource cost

Table 3 gives flash and RAM consumption as reported by the compiler for both framework versions. The security features added in the hardened configuration cost approximately 3 kB of flash and 170 bytes of RAM. We emphasise that this is a static measure: no CPU-time or energy cost has been measured, and both remain open.

Table 3: Build cost, as reported by the compiler in continuous integration. Static sizes only; no run-time or energy cost has been measured.

Framework version	Flash (B)	of budget	RAM (B)	of budget
arduino-esp32 2.0.17	793,433	60.5%	46,372	14.2%
arduino-esp32 3.3.12	968,221	73.9%	49,876	15.2%

8 Experimental Methodology

8.1 Evidence categories

Results in this paper fall into three categories and we label each one, because they are not interchangeable. *Formal verification results* come from exhaustive model checking. *Simulation results* come from the unmodified firmware driven by a seeded harness on a host. *Analytical results* follow from the construction, such as the 2^{-128} forgery probability. No result in this paper is a physical hardware measurement; the single exception is the build cost of Table 3, which a real compiler measured but which describes the binary rather than the running device.

8.2 Hypotheses

H1 (logical isolation). No execution of the system exists in which network activity blocks, delays or corrupts the safety path, and no forged, stale or malformed command is accepted. *Testable by model checking and simulation.*

H2 (timing isolation). $F_\Lambda \stackrel{d}{=} F_0$ for radio-layer load Λ across the operating range. *Requires hardware; not tested here.*

Given [1], we do not expect H2 to hold exactly. The informative question is the magnitude of any shift relative to the deadline margin, and whether queue discipline bounds it.

8.3 Determinism and reproduction

Nothing in the evaluation is collected as a dataset. Inputs are generated deterministically from fixed seeds, which is stronger for this purpose than distributing a file: an independent researcher regenerates the identical inputs by running the scripts. Each trial of the adversarial campaign and the ablation executes in a forked child process, because the firmware's globals are file-scope and would otherwise leak between trials. This was not a stylistic choice: before it was introduced, state leakage produced a spurious invariation violation at zero attack rate.

The campaign and fuzzing pipeline has been verified to regenerate its summary files byte-identically from the committed seeds.

8.4 Metrics

8.5 Experimental configurations

The adversarial campaign runs 3,000 trials per build across six attack rates from 0 to 20,000 messages/s. The adversary injects forged tags signed with a key it does not hold, replays of messages captured on the wire, bare verbs carrying no credential, and random bytes. The ablation runs 300 trials per design over the same rates with identical seeds. The fuzzer runs 3,000,000 iterations over six input classes under AddressSanitizer and UndefinedBehaviorSanitizer. The sensitivity sweep runs 2,000 benign trials of ten simulated minutes per grid cell.

Table 4: Evaluation metrics and their definitions.

Metric	Definition
Safety latency	Simulated milliseconds from confirmed sensor assertion to entry-delay entry
Command availability	Trials in which a legitimate command issued during the flood took effect
Integrity failure	Trials in which an adversary's message caused a state change
Replay defeat	Trials in which the adversary obtained an accepted command using only captured or forged material
Invariant violation	Trials violating any safety invariant
Accepted fuzz input	Inputs changing state or advancing the counter without a valid tag
False-positive rate	Benign runs in which the detector reported at least once
Detection latency	Simulated milliseconds from attack onset to first detector report

Table 5: Command-path ablation. 300 trials per design, identical adversary and seeds. "Survives flood" counts flooded trials in which a legitimate command took effect; "defeated by replay" counts trials in which the adversary obtained an accepted command using only captured or forged material.

Design	Survives flood	Defeated by replay	Alarm fired
No command defence	0/250	250/300	300/300
Rate limit, 5 commands/s	0/250	246/300	300/300
Static shared token	250/250	250/300	300/300
HMAC-SHA256 + counter	**250/250**	**0/300**	**300/300**

9 Results

9.1 Adversarial campaign

Across 6,000 trials no adversary disarmed the device and no safety invariant was violated. Without command authentication the adversary changed the system state in all 2,500 flooded trials; with authentication, in none. The availability result is the more informative one: with authentication a legitimate command took effect in all 3,000 trials regardless of attack rate, while in the baseline flooding starved it in every flooded trial.

We report no hypothesis test on these data. Because the harness advances a logical clock there is no contention, the safety latency is a deterministic 151 ms in every trial—the sensor confirmation window plus loop period—and its variance is identically zero. Any test on two constant-valued samples is trivially satisfied and carries no information. We pre-register the analysis for the hardware measurements instead: a two-sided Mann–Whitney U test with tie correction, chosen because latency distributions on a real-time operating system are not normal, with the implementation committed in advance.

9.2 Command-path ablation

To avoid resting on a baseline that is defeated by construction, we ran the same adversary against four designs: no command defence, a token-bucket rate limiter admitting five commands per second, a static shared token carried in the payload, and the signed-counter path. The rate limiter and the token are modelled as edge filters, which is where a developer would place them in practice. Table 5 reports the outcome.

The result separates two properties that are easily conflated. A rate limiter cannot distinguish a legitimate command from a flood, so it discards both and sacrifices availability while leaving integrity entirely untouched. A static shared token *does* restore availability, because the edge filters the flood cheaply—but it travels in plaintext, so a single captured command defeats it. Only the monotonic counter prevents replay; a message authentication code alone would behave as the token row does. The practical

Table 6: Detector sensitivity. False-positive rate is measured at 60 benign frames per minute; detection latency is the median at the stated attack rate. The deployed configuration is shown in bold.

Threshold	Window (ms)	FP rate	Detect, 20 f/s	Detect, 5 f/s	Missed, 5 f/s
5	2000	1.0000	234 ms	948 ms	0/500
8	1000	0.0355	405 ms	2326 ms	0/500
8	2000	0.7985	383 ms	1520 ms	0/500
12	**2000**	**0.0025**	**597 ms**	**2673 ms**	**0/500**
12	5000	0.9450	577 ms	2348 ms	0/500
20	2000	0.0000	1004 ms	26137 ms	279/500
20	5000	0.0005	958 ms	3852 ms	0/500

implication is that a designer evaluating a command-path defence against availability alone will accept a design that is trivially broken on integrity.

The alarm fired in every trial of every design. This is the isolation property behaving as intended: the safety function does not depend on the command path, so no command-path design choice affects whether the siren sounds.

9.3 Detector parameter sensitivity

The detector's threshold and window had a stated rationale but no measurement. Sweeping them against a Poisson model of benign management-frame arrivals showed that our initial threshold of eight was wrong. Access points emit deauthentication and disassociation frames in normal operation during roaming, idle timeout, load balancing and reboot; at 60 frames per minute, representing a busy multi-access-point environment, a threshold of eight fired spuriously in 79.9% of ten-minute runs. Table 6 gives the trade-off surface.

We revised the firmware default to twelve, which reduces the false-positive rate to 0.25% while still detecting every attack rate swept with no misses, at a cost of approximately 214 ms of additional detection latency at 20 frames per second. A detector that reports frequently on a healthy network trains its owner to ignore it, and on this device each report consumes an event-queue slot; the additional latency is a reasonable price.

We are explicit that the benign arrival model is an *assumption*, not a measurement. No capture of real management-frame traffic was available to us. These results therefore characterise the detector's logic against a stated model and are not a false-positive rate for any real network. Obtaining one requires a passive capture on a production network and is future work.

9.4 Fuzzing

The command parser is the one input a network peer fully controls. Across 3,000,000 iterations under AddressSanitizer and UndefinedBehaviorSanitizer the fuzzer found no memory error, no undefined behaviour, and no input that changed state or advanced the counter. We read this as evidence of memory safety and parser robustness rather than of authentication soundness: against a 128-bit tag, zero forgeries is the expected outcome and is not independent evidence.

One episode is worth recording as a methodological caution. An early run reported an accepted input. Investigation showed the mutation had flipped the case of a hexadecimal digit, which decodes to identical tag bytes, so the input was the genuine command and acceptance was correct behaviour. The corpus now signs its mutation seed with a deliberately incorrect key so that no mutation can produce a valid command by accident. Had this not been chased down it would have become a false claim.

10 Independent Hardware Validation Framework

The results above establish H1. They cannot establish H2, because the simulated clock does not model contention for the bus, the cache or the interrupt controller, and that contention is precisely the phenomenon in question.

We did not have access to the required hardware. Rather than leave H2 as an unqualified gap, we treat the validation apparatus itself as a deliverable: firmware for the measuring instrument, wiring, experimental conditions, collection and analysis scripts, pre-registered statistics and explicit falsification criteria, all public, so that any laboratory with two inexpensive boards can perform the measurement and report it. The decomposition of Section 5 is what makes the result interpretable when they do, since it establishes that a measured shift can be attributed to $I_{\mathrm{isr}} + I_{\mathrm{mem}}$ and to nothing else.

We state plainly that this is a framework and not a result. No physical measurement appears in this paper.

10.1 Apparatus

Figure 3 shows the arrangement. The device under test runs the unmodified firmware. A second, independent microcontroller acts as the measuring instrument: it drives the sensor line, timestamps that edge with its own hardware timer, and timestamps the rising edge of the siren line. The measuring clock is therefore never the clock under load, which is essential: a device cannot credibly measure its own latency while that latency is the quantity being perturbed.

10.2 Conditions

Four conditions, each compared against the unloaded control:

A. Idle. No network activity. Establishes F_0 and the instrument's own noise floor.

B. Legitimate load. Normal MQTT traffic with the broker reachable.

C. Command-interface stress. High-rate publication on the command topic.

D. Radio-layer stress. Deauthentication and disassociation frames at the MAC layer.

Conditions A through C require only the two boards and a probe. Condition D requires an authorised isolated or shielded radio environment and institutional permission, and is the condition that distinguishes this work from prior IP-layer measurements.

10.3 Analysis plan, pre-registered

Each loaded condition is compared against condition A using a two-sided Mann–Whitney U test with tie correction. For samples of size n_1 and n_2 with rank sum R_1,

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}, \qquad U = \min(U_1,\, n_1 n_2 - U_1), \tag{10}$$

$$\sigma_U^2 = \frac{n_1 n_2}{12}\left[(n+1) - \frac{\sum_i (t_i^3 - t_i)}{n(n-1)}\right], \qquad n = n_1 + n_2, \tag{11}$$

with $z = (U - n_1 n_2/2)/\sigma_U$ and a two-sided p-value from the normal approximation. The test is non-parametric because latency distributions on a real-time operating system are not normal. The bracketed term in (11) is the tie correction, which matters here because quantised timer readings produce many ties. The implementation is committed in advance of any measurement, and we will report effect size and the full distribution rather than a p-value alone.

10.4 Ethics

The radio-layer condition is to be performed only on equipment owned by the experimenter, in an isolated or shielded environment, with institutional permission obtained beforehand. No third-party network or device is targeted. Neither this paper nor its public repository contains any attack-generation procedure or tooling. We note that deauthentication tooling for this chip family is already published [8, 9]; we cite that work as evidence the threat is realistic while declining to add to it.

11 Discussion

11.1 What the evidence supports

The formal and simulation results together establish that the design contains no path by which network activity reaches the safety function logically. The TLA+ isolation model is the stronger of the two, because granting the network task no fairness means the property holds under every possible interleaving rather than the ones a harness happens to generate. The campaign and fuzzing results confirm that the implementation matches the specification.

The ablation is the result we consider most likely to be useful to other designers, because it is the least predictable. It is tempting to evaluate a command-path defence by asking whether legitimate commands survive an attack. Both the static token and the signed-counter design pass that test identically, and a designer stopping there would conclude they are equivalent. They are not: one is defeated by a passive eavesdropper, and the measurement that distinguishes them is capture-and-replay rather than throughput.

11.2 Why the design behaves as observed

The availability result follows from where verification happens rather than from cryptography. Because the tag is checked on the network core before the bounded command queue, a flood of invalid commands is discarded without ever occupying a queue slot. The queue therefore retains capacity for the one valid command, and admission is decided by validity rather than arrival order. A rate limiter inverts this: it decides by arrival order, which an attacker controls.

11.3 Relationship to prior measurement

Our work complements rather than contradicts [1]. They established that on-chip network activity reaches a critical task on this hardware even with cross-core separation; we accept that result and address the question it leaves open, namely whether the mitigations available on stock hardware are adequate, under the load class that actually threatens a wireless alarm. Their proposed mitigations require either hardware that does not exist [2] or a platform an order of magnitude larger [3]. If our pending measurement shows that pinning with queue discipline holds the deadline under radio-layer load, that is a useful positive result for a very large installed base. If it shows the opposite, that is a more important result still, because it would indicate that the standard advice given to developers of such devices is insufficient and that the installed base is exposed.

12 Threats to Validity

Construct validity. This is the most serious threat. The dependent variable in simulation is a logical latency measured on a clock the harness itself advances. We call it safety latency, but it is not the physical quantity of interest, and the divergence between the construct and the measure is the reason Section 10 exists.

Internal validity. Largely controlled. Seeds are fixed, the pipeline regenerates byte-identically, and continuous integration re-runs a reduced version on every commit. Per-trial process isolation was introduced specifically because global state leakage had produced a spurious violation. Two further harness defects were found during the ablation and are documented: the adversary could act during the experimenter's own control actions, and could leave the system mid-transition so that the intrusion phase was skipped, which made the weaker designs appear to fail in a way they had not. The residual threat is that the host models and the firmware share an author, so a common misconception would be invisible to both.

External validity. Limited, and we scope our claims accordingly. All results concern one chip, two framework versions, one application and no deployment. We do not generalise to single-chip Wi-Fi SoCs as a class. The single-core variants of the same family are arguably the more interesting case, since they cannot pin at all.

Conclusion validity. We report no statistical conclusion from the simulation, precisely because the zero-variance data cannot support one. The formal conclusions are sound within their abstraction, and that abstraction excludes timing. The security conclusions of the campaign follow deductively from the construction and are therefore valid but uninformative; the ablation was added because of this.

Bias. There is no blinding, and the analysis author is the implementation author. For the hardware campaign this matters more than it does in simulation, because expectancy effects can influence setup and outlier handling. The measurement protocol and the analysis script are committed before any data is collected, which is a pre-registration in substance.

13 Limitations and Future Work

The principal limitation is stated plainly: no physical measurement has been performed, and the timing-invariance claim is consequently untested. No latency figure appears in this paper that we have not measured, and none is estimated.

Beyond this: the benign arrival model underlying the sensitivity analysis is synthetic; the per-command CPU cost of tag verification and the energy cost of both the security features and promiscuous-mode monitoring are unmeasured; the ablation lacks an arm representing transport security with broker credentials; no ablation isolates the contribution of each of the four security mechanisms individually; flash encryption, secure boot and signed over-the-air update are not implemented; and the false-alarm rate of the sensor-fusion modes has not been characterised against real sensors.

The ordered priorities for future work are therefore: conditions A through C of the hardware protocol, which need only two boards and a probe; condition D, subject to approval; a passive capture to replace the synthetic arrival model; and replication on a single-core variant of the same family.

14 Conclusion

We treated a consumer intrusion alarm as a safety-critical system and asked whether the one isolation mitigation available on its hardware—core pinning, combined with bounded queue discipline—is adequate when the attacked resource is the chip's own radio. Prior measurement establishes that network activity reaches a critical task on this hardware despite cross-core separation; what it does not establish is whether the available mitigation suffices, and it has not examined radio-layer load, an adversarial model, or any formal treatment.

We answer the logical half of the question with exhaustive model checking under a network task granted no fairness, with 7,200 seeded adversarial trials and with three million sanitizer-backed fuzzing iterations. We show through ablation that availability and integrity are separable properties and that a

plausible defence can satisfy the first while failing the second completely. We show through sensitivity analysis that our own initial detector threshold was wrong by a wide margin, and we corrected it. For the timing half we contribute an analysis and an apparatus rather than a number: a decomposition that removes three of four interference terms by construction and names the one that remains, and a public framework with which an independent laboratory can measure it. We report no latency figure, because we have not measured one.

All claims are scoped to the platform tested: the dual-core Xtensa ESP32 under arduino-esp32 2.0.17 and 3.3.12. The artifact is public so that the results can be reproduced and the open measurement completed, by us or by others.

Declarations

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Conflicts of interest. The author declares no conflict of interest.
Author contributions. MD Humayun: conceptualisation, methodology, software, formal analysis, investigation, data curation, writing (original draft, review and editing), visualisation.
Data availability. All data reported here are generated deterministically from fixed seeds by scripts in the public repository. Aggregate results and the raw per-trial records for the ablation and sensitivity experiments are committed; the campaign's per-trial records regenerate identically by running the provided script. No third-party or human-subject data were used.
Code availability. Firmware, formal specifications, experiment harnesses, analysis scripts and figure sources are publicly available under the MIT licence. Continuous integration re-runs the formal verification and a reduced adversarial campaign on every commit.
Ethics. No human or animal subjects were involved, and no third-party network, device or system was attacked or tested. The repository publishes no attack-generation procedure or tooling.

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Use of AI-assisted tools. An AI assistant was used for drafting assistance, documentation, code review and literature discovery. All research claims, experimental designs, results and conclusions are the author's own. Every result reported here was produced by the committed scripts and verified by the author, and every reference was checked against its publisher or preprint record.

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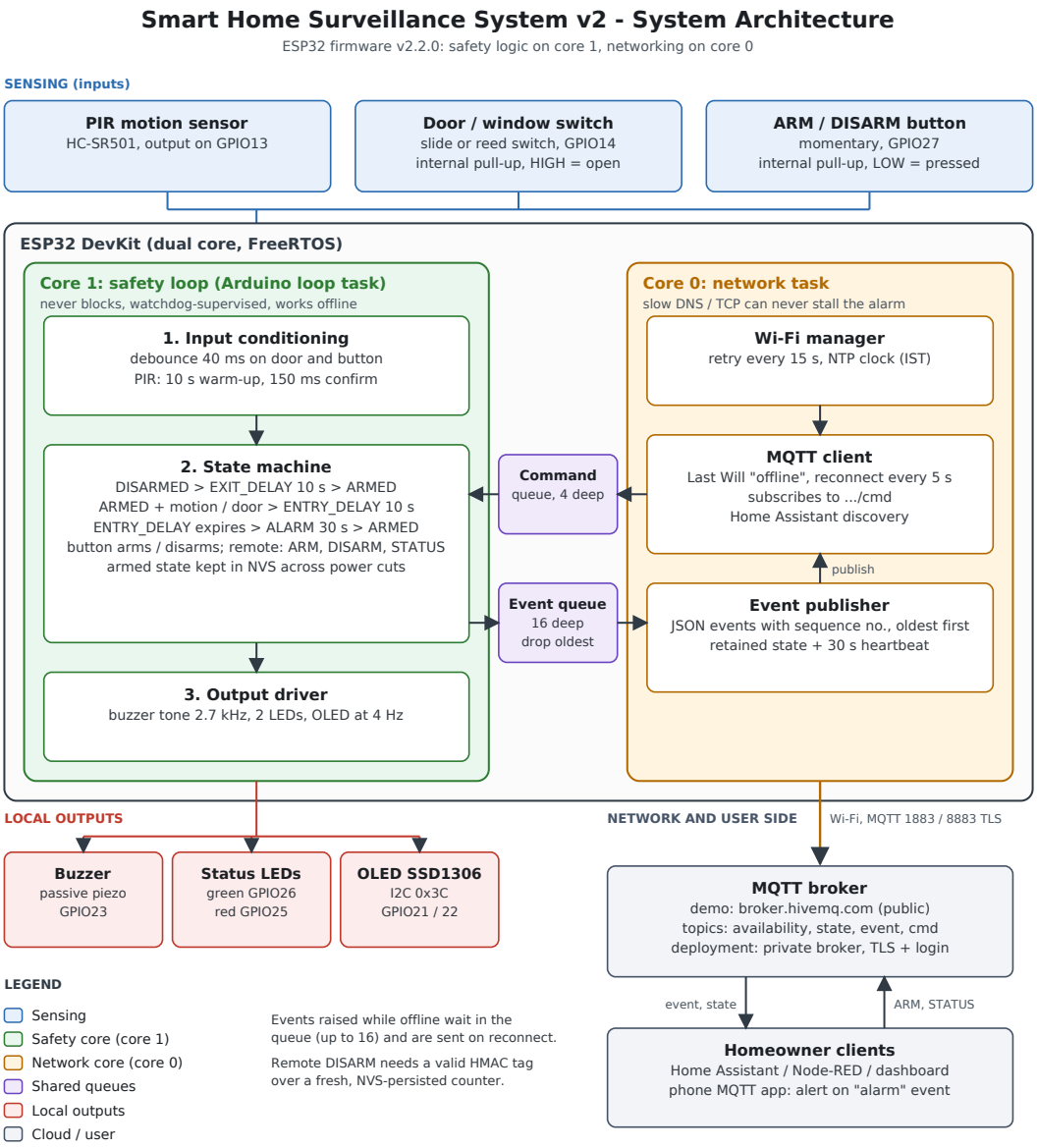


Figure 1: System architecture. The safety domain on core 1 owns the state machine, sensing and actuation; the network domain on core 0 owns Wi-Fi, MQTT and command authentication. The two communicate only through two bounded queues and a status snapshot guarded by a zero-timeout mutex.

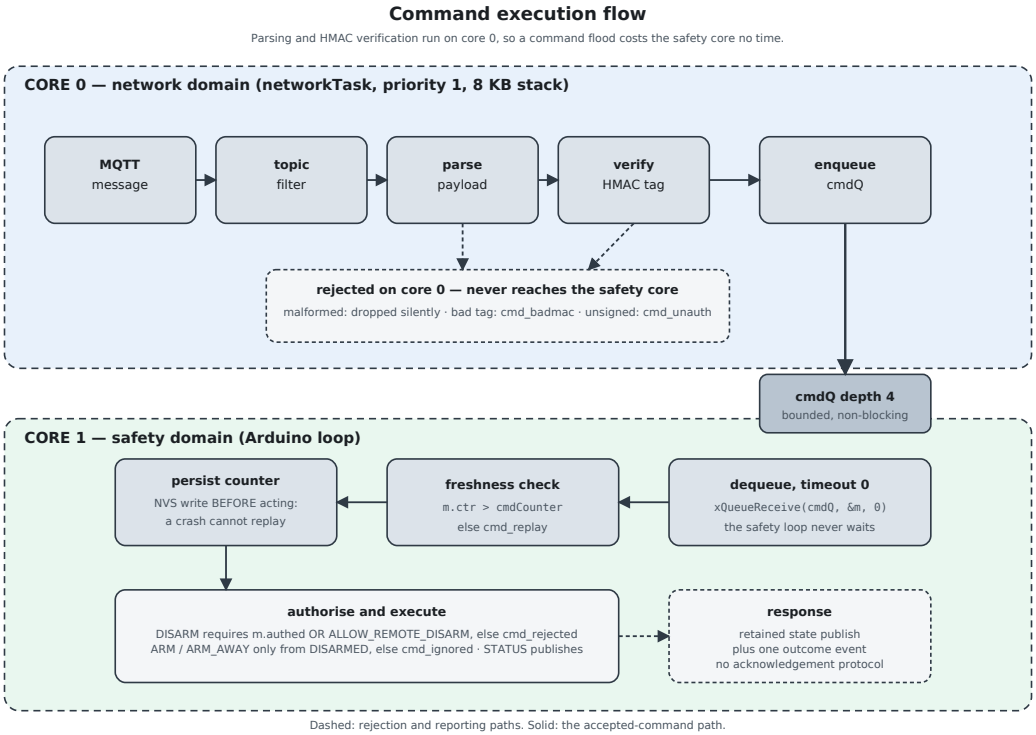


Figure 2: Command execution across the two cores. Parsing and tag verification complete on core 0, so a command flood costs the safety core no time; the replay counter is committed to non-volatile storage before the command is acted upon.

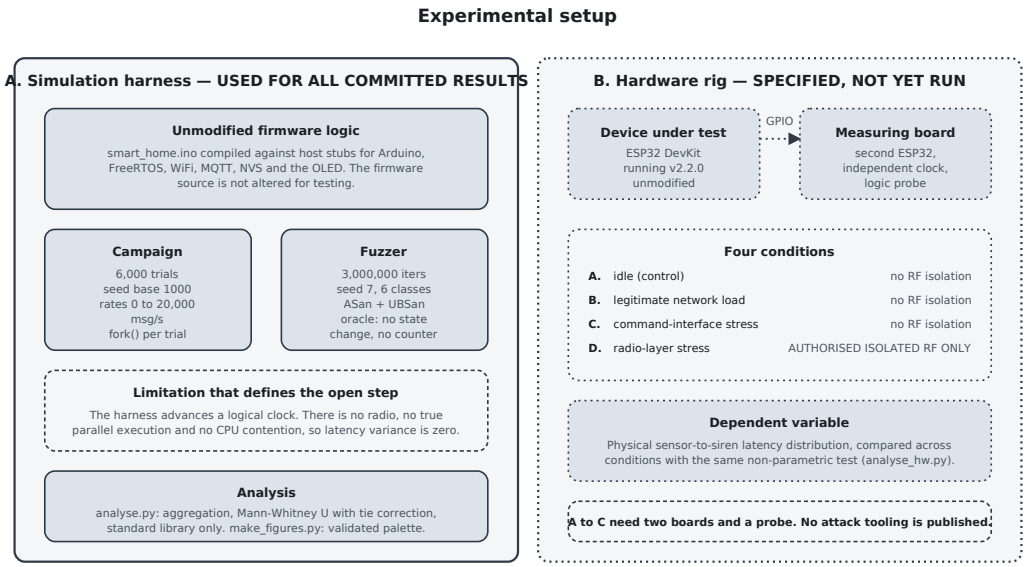


Figure 3: Experimental setup. (A) The simulation harness used for all results reported in this paper. (B) The hardware rig, specified and built but not yet run, in which a second microcontroller serves as an independent measuring instrument across four load conditions.